CS336 · AIML · B.Tech VI Sem CSE-C

AI Agent for Disease Risk Awareness and Prevention

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Assignment 3:- https://github.com/Nischal07bot/disease_awarness_ai_agent

Problem Statement & Target Audience

Problem Statement

- Early detection of Diabetes (PIMA dataset) and Heart Disease (UCI dataset) is critical to prevent severe health complications.
- Traditional diagnosis:
  Requires multiple clinical tests
  Can be time-consuming and expensive
- There is a need for a data-driven predictive system that:
  Uses patient health parameters
  Provides fast, reliable risk prediction
- This project aims to build ML models that can:
  Predict likelihood of Diabetes and Heart Disease
  Assist in early screening and decision support

Clinical Decision Support

Rural/Remote Health Screening

Personal Health Monitoring

Target Audience

👨‍⚕️ Healthcare Professionals
Doctors for preliminary screening support

🏥 Hospitals & Clinics
Assist patients

👩‍💻 Researchers & Data Scientists
For experimentation on medical datasets

🧑‍🤝‍🧑 General Public (Future Scope)
Awareness tools for early risk detection

Dataset Description

PIMA Indians Diabetes

Samples	768 (8 features + target)
Target	Diabetic (1) / Non-Diabetic (0)
Class Ratio	1.87 : 1
Preprocessing	436 outliers (IQR): There are many outliers, therefore, they are not removed and Robust scaler is applied on the columns; 80/20 split → Train 614 / Test 154

Features: Pregnancies · Glucose · BloodPressure · SkinThickness · Insulin · BMI · DiabetesPedigreeFunction · Age

UCI Heart Disease

Samples	303 (13 features + target)
Target	Disease Present (1) / Absent (0)
Class Ratio	1.17 : 1 (near-balanced)
Preprocessing	85 Outliers:There are many outliers, therefore, Robust Scaling applied; 80/20 split → Train 237 / Test 60

Features: Age · Sex · Chest Pain · Resting BP · Cholesterol · Fasting BS · ECG · Max HR · Exang · Oldpeak · Slope · CA · Thal

Model Architecture — AI Disease Risk Agent

Preprocessing (NB1) → **Feature Engg. (NB2)** → **Model Training (NB3)** → **Explainability (NB4)** → **AI Agent (NB5)**

Machine Learning Pipeline & Model Overview

- Preprocessing: Imputes missing health values (Insulin, SkinThickness); applies RobustScaler for fair model training
- Feature Engg.: Creates domain-specific risk indicators (Glucose_Insulin_Ratio, chol_bp_ratio); MI scoring selects most disease-relevant features
- Model Training: Trains LR, RF, GB with GridSearchCV; outputs risk probability (0–1) per patient;

- Model Evaluation:
  Accuracy, Precision, Recall, F1-score
  ROC-AUC Curve
- Prediction System:
  Input patient data → Output risk prediction

Results & Interpretation

0.8889
Best PIMA Recall
(Logistic Regression)

0.8214
Best Heart Recall
(Logistic Regression)

0.7338
PIMA Accuracy

0.8214
Heart Accuracy

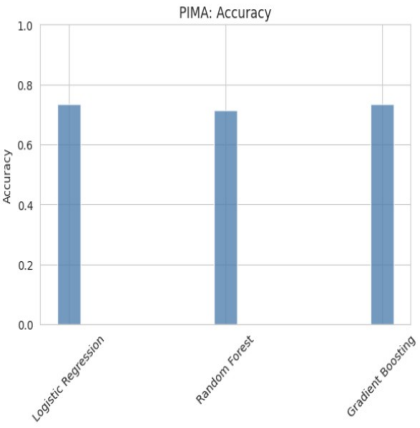
Model Performance on Test Set

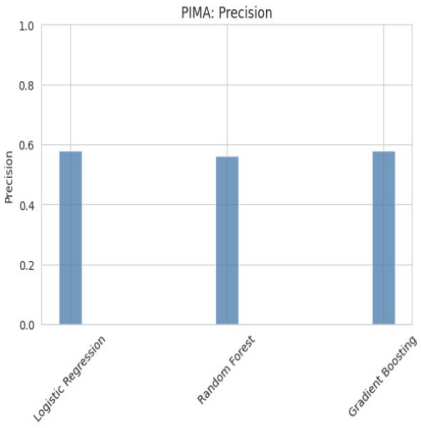
Model	PIMA Acc	PIMA F1	PIMA AUC	Heart Acc	Heart F1	Heart AUC
Logistic Regression	0.7338	0.7007	0.8276	0.8500	0.8364	0.9420
Random Forest	0.7143	0.6765	0.8191	0.8333	0.8077	0.9342
Gradient Boosting	0.7338	0.6963	0.8157	0.8167	0.7843	0.9275

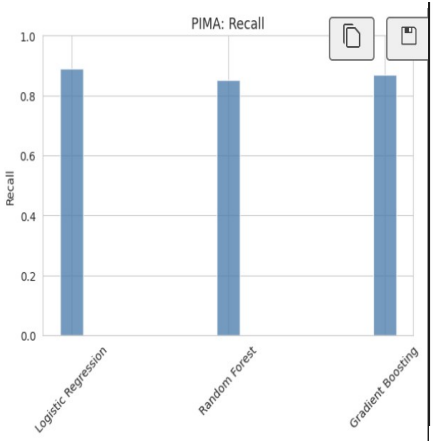
Key Observations

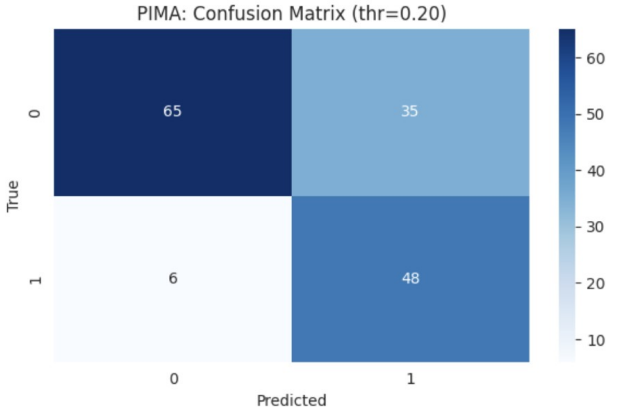
- Linear Regression best on both datasets
- Top PIMA features (MI): Glucose 0.136, BMI 0.0911
- Top Heart features (MI): thal 0.143, cp 0.115

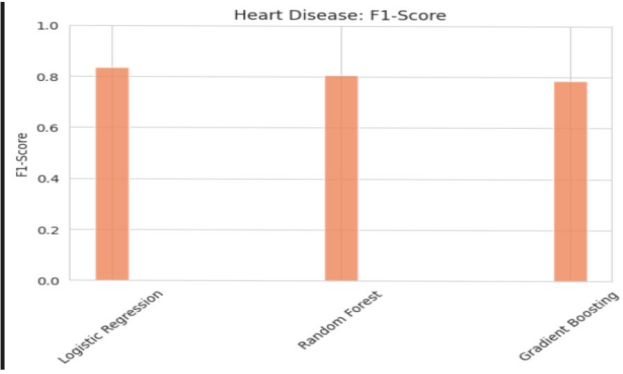
Graphs & Visualization

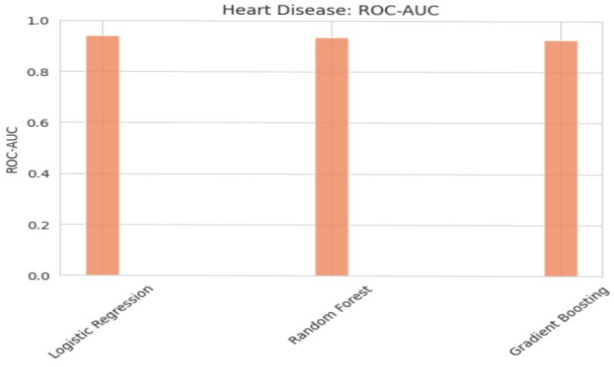


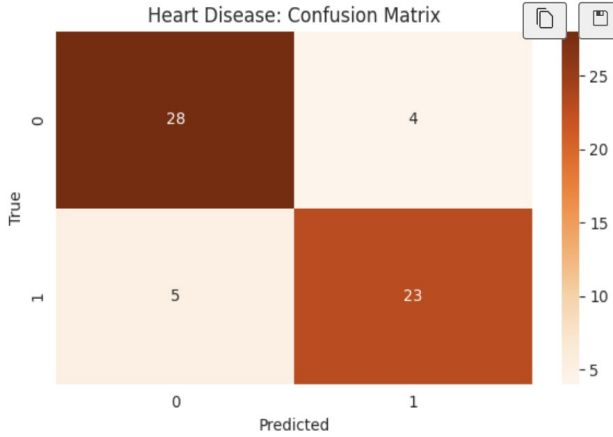












AI Agent in Action & Conclusion

DiseaseRiskAgent Logic

Low Risk (prob < 0.3)

"Maintain lifestyle, annual checkup"

Medium Risk (0.3 – 0.7)

"200 min/week exercise, reduce sugar, quarterly checkups"

High Risk (prob > 0.7)

"Urgent consultation with healthcare provider"

Sample Agent Outputs (NB5):

Prob 0.218 → Low Risk → Lifestyle maintenance

Prob 0.374 → Medium Risk → Dietary + monitoring

Prob 0.401 → Medium Risk → Lifestyle changes (Heart)

Conclusion

✅ Successfully built a 5-notebook end-to-end AI agent for disease risk prediction

✅ Real fairness metrics computed — age-group bias identified and documented

✅ System is explainable (SHAP + LIME) and ethically analysed

⚠️ Moderate accuracy due to synthetic dataset — real data would improve performance significantly

✅ Deployment-ready with documented limitations, encryption, and user rights